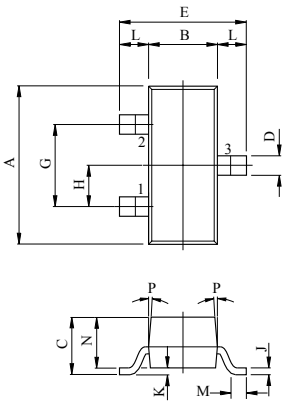
KTK5162S

N CHANNEL MOS FIELD EFFECT TRANSISTOR

ULTRA-HIGH SPEED SWITCHING APPLICATIONS
ANALOG SWITCH APPLICATIONS

EATURES

- 2.5 Gate Drive.
- Low Threshold Voltage : V_{th}=0.5～1.5V.
- High Speed.
- Small Package.
- Enhancement-Mode.



DIM	MILLIMETERS
A	2.93±0.20
B	1.30+0.20/-0.15
C	1.30 MAX
D	0.45+0.15/-0.05
E	2.40+0.30/-0.20
G	1.90
H	0.95
J	0.13+0.10/-0.05
K	0.00 ~ 0.10
L	0.55
M	0.20 MIN
N	1.00+0.20/-0.10
P	7°

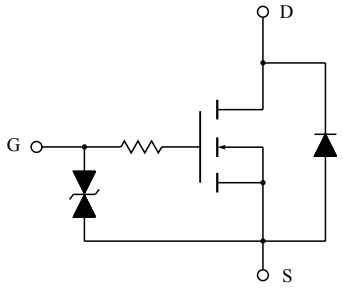
1. SOURCE
2. GATE
3. DRAIN

SOT-23

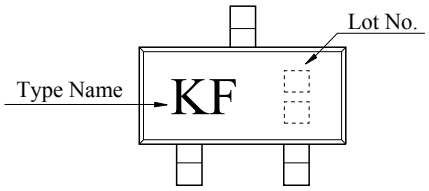
MAXIMUM RATINGS (Ta=25℃)

CHARACTERISTIC	SYMBOL	RATING	UNIT
Drain-Source Voltage	V_{DS}	60	V
Gate-Source Voltage	V_{GSS}	±20	V
DC Drain Current	I_D	100	mA
Drain Power Dissipation	P_D	200	mW
Channel Temperature	T_{ch}	150	℃
Storage Temperature Range	T_{stg}	-55～150	℃

EQUIVALENT CIRCUIT



Marking

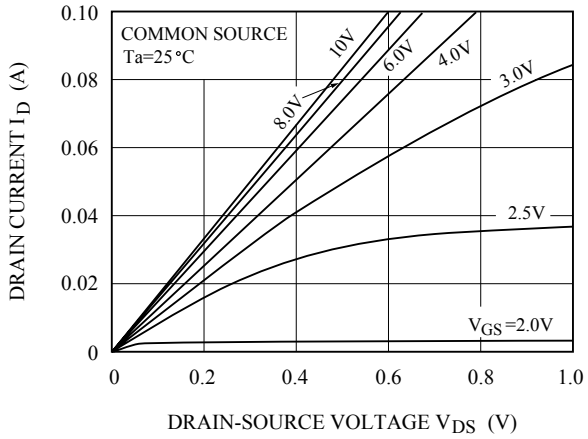


THIS TRANSISTOR IS ELECTROSTATIC SENSITIVE DEVICE.
PLEASE HANDLE WITH CAUTION.

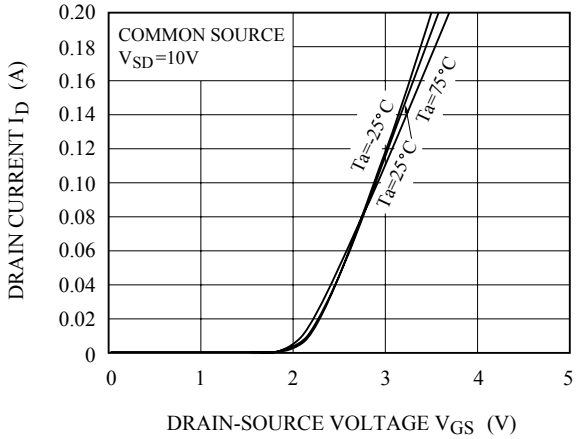
ELECTRICAL CHARACTERISTICS (Ta=25℃)

CHARACTERISTIC		SYMBOL	TEST CONDITION	MIN.	TYP.	MAX.	UNIT
Gate Leakage Current		I_{GSS}	V_{GS}=±16V, V_{DS}=0V	-	-	±1	μA
Drain-Source Breakdown Voltage		$V_{(BR)DSS}$	I_D=100μA, V_{GS}=0V	60	-	-	V
Drain Cut-off Current		I_{DSS}	V_{DS}=60V, V_{GS}=0V	-	-	1	μA
Gate Threshold Voltage		V_{th}	V_{DS}=3V, I_D=0.1mA	0.5	-	1.5	V
Forward Transfer Admittance		$\|Y_{fs}\|$	V_{DS}=10V, I_D=50mA	85	120	-	mS
Drain-Source ON Resistance		$R_{DS(ON)}$	I_D=10mA, V_{GS}=2.5V	-	22	40	Ω
Input Capacitance		C_{iss}	V_{DS}=10V, V_{GS}=0V, f=1MHz	-	6.2	-	pF
Reverse Transfer Capacitance		C_{rss}	V_{DS}=10V, V_{GS}=0V, f=1MHz	-	1.5	-	pF
Output Capacitance		C_{oss}	V_{DS}=10V, V_{GS}=0V, f=1MHz	-	4.4	-	pF
Switching Time	Turn-on Time	t_{on}	V_{DD}=25V, I_D=50mA, V_{GS}=0～2.5V	-	10	-	nS
	Turn-off Time	t_{off}		-	105	-	nS

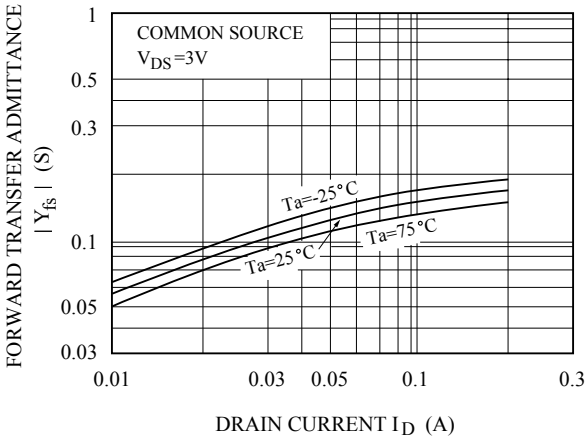
I_D - V_{DS}



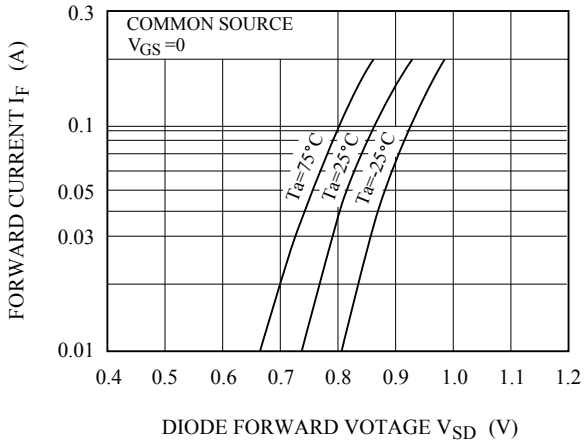
I_D - V_{GS}
(LOW VOLTAGE REGION)



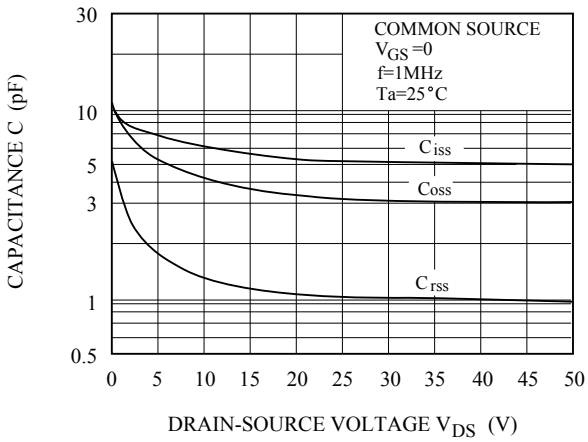
$|Y_{fs}|$ - I_D



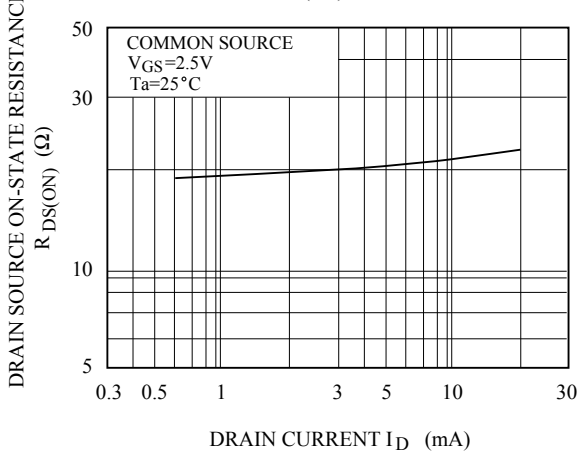
I_F - V_{SD}



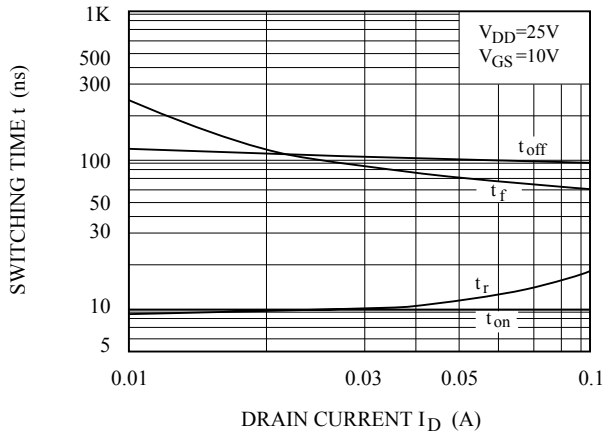
C - V_{DS}



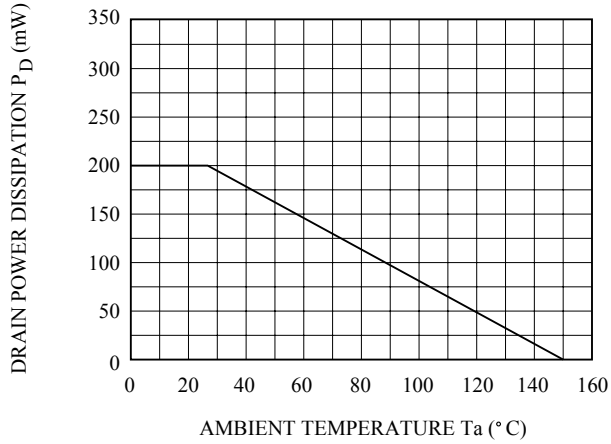
$R_{DS(ON)}$ - I_D



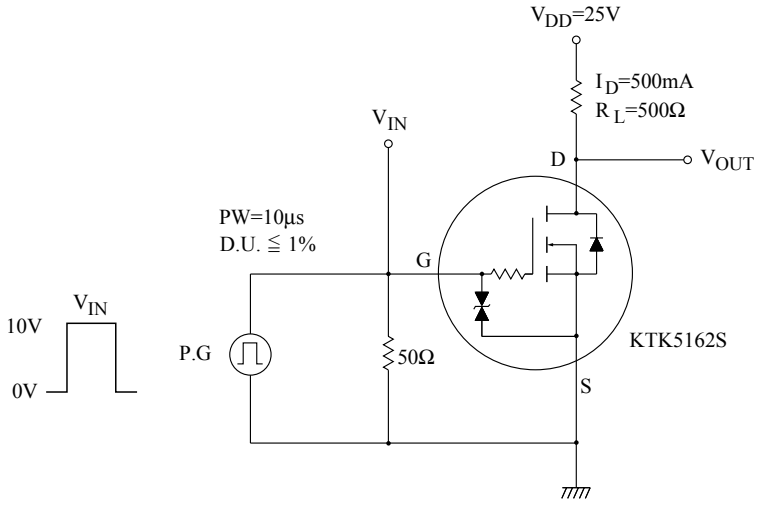
t - I_D



P_D - Ta



SWITCHING TIME TEST CIRCUIT



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Datasheets for electronics components.